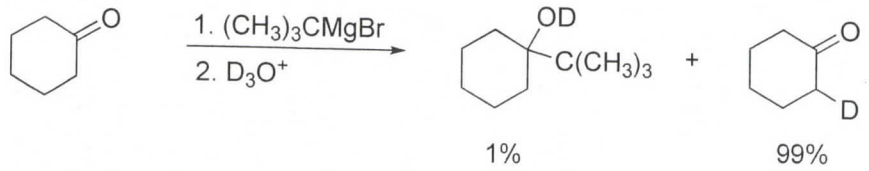
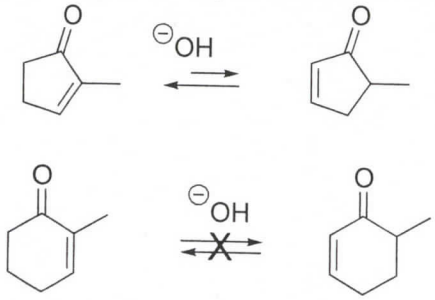
Organic Cume
February 15, 2019

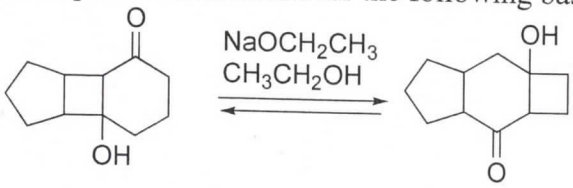
1. Attempted Grignard reaction of cyclohexanone with *tert*-butylmagnesium bromide yields only about 1% of the expected addition product along with 99% unreacted cyclohexanone. If D_3O^+ is added to the reaction mixture after a suitable period, however, the "unreacted" cyclohexanone is found to have one deuterium atom incorporated into it. Explain.



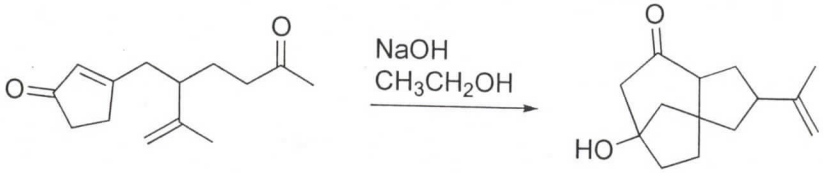
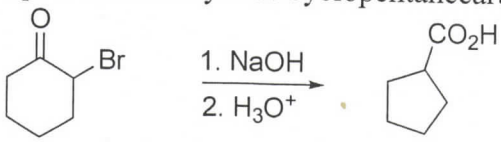
2. 2-Substituted 2-clopentenone can be interconverted with 5-Substituted 2-clopentenone in the presence of hydroxide. However, the analogous isomerization is not observed for 2-substituted 2-cyclohexenone. Explain.



3. Propose a mechanism for the following base-catalyzed isomerization.



4. Propose a mechanism for the following reaction.



5. The Favorskii reaction involves treatment of an alpha-bromo ketone with base to yield a ring-contracted product. For example, reaction of 2-bromo-cyclohexanone with aqueous NaOH yields cyclopentanecarboxylic acid. Propose a mechanism.